

Problem 13

Find the random event X when:

$$\overline{X + A} + \overline{X + \bar{A}} = B$$

Solution

We will use De Morgan's laws and Boolean algebra to solve for X .

Step 1: Apply De Morgan's Law

Recall De Morgan's law: $\overline{X + Y} = \bar{X}\bar{Y}$

$$\begin{aligned}\overline{X + A} &= \bar{X}\bar{A} \\ \overline{X + \bar{A}} &= \bar{X}\bar{\bar{A}} = \bar{X}A\end{aligned}$$

Step 2: Substitute into the Equation

$$\bar{X}\bar{A} + \bar{X}A = B$$

Step 3: Factor Out $\bar{X}$

$$\bar{X}(\bar{A} + A) = B$$

Step 4: Simplify Using Complement Law

Since $\bar{A} + A = S$ (the certain event):

$$\begin{aligned}\bar{X} \cdot S &= B \\ \bar{X} &= B\end{aligned}$$

Step 5: Solve for X

Taking the complement of both sides:

$$X = \bar{B}$$

Verification

Let's verify that $X = \bar{B}$ satisfies the original equation.

Left side:

$$\overline{X + A} + \overline{X + \bar{A}} = \overline{\bar{B} + A} + \overline{\bar{B} + \bar{A}}$$

First term:

$$\overline{\bar{B} + A} = \bar{\bar{B}} \cdot \bar{A} = B\bar{A}$$

Second term:

$$\overline{\bar{B} + \bar{A}} = \bar{\bar{B}} \cdot \bar{\bar{A}} = BA$$

Sum:

$$B\bar{A} + BA = B(\bar{A} + A) = B \cdot S = B$$

Indeed, the left side equals B , which matches the right side.

Alternative Verification Using Original Substitution

With $X = \bar{B}$:

$$\overline{X + A} = \overline{\bar{B} + A}$$

Using De Morgan's law:

$$= B\bar{A}$$

$$\overline{X + \bar{A}} = \overline{\bar{B} + \bar{A}}$$

Using De Morgan's law:

$$= BA$$

Therefore:

$$\overline{X + A} + \overline{X + \bar{A}} = B\bar{A} + BA = B(\bar{A} + A) = B \cdot S = B$$

Uniqueness

Is this solution unique? Let's check if there could be other solutions.

From our derivation, we had:

$$\bar{X} = B$$

This directly gives $X = \bar{B}$, which is the unique solution.

Interpretation

The equation $\overline{X + A} + \overline{X + \bar{A}} = B$ can be interpreted as:

- "X or A doesn't occur" OR "X or not-A doesn't occur" equals "B occurs"
- Equivalently: "(neither X nor A) or (neither X nor not-A)" equals "B occurs"
- This simplifies to: "not-X" equals "B occurs"
- Therefore: X is the complement of B

Special Cases

Case 1: If $B = S$ (certain event), then $X = \bar{S} = \emptyset$ (impossible event).

Case 2: If $B = \emptyset$ (impossible event), then $X = \bar{\emptyset} = S$ (certain event).

Case 3: If A is irrelevant to the problem, the equation still holds as $X = \bar{B}$ regardless of A.

Probabilistic Interpretation

If we assign probabilities:

$$P(\overline{X + A}) + P(\overline{X + \bar{A}}) = P(B)$$

Note that in general:

$$P(\overline{X + A}) + P(\overline{X + \bar{A}}) = P(\bar{X}\bar{A}) + P(\bar{X}A) = P(\bar{X})$$

assuming the events are mutually exclusive (which they are since $\bar{A}$ and A are disjoint).

So we need $P(\bar{X}) = P(B)$, which for the events themselves means $X = \bar{B}$.

Final Answer

$$\boxed{X = \bar{B}}$$

Derivation summary:

1. Apply De Morgan's law: $\overline{X + A} = \bar{X}\bar{A}$ and $\overline{X + \bar{A}} = \bar{X}A$
2. Substitute: $\bar{X}\bar{A} + \bar{X}A = B$
3. Factor: $\bar{X}(\bar{A} + A) = B$
4. Simplify: $\bar{X} \cdot S = B$, so $\bar{X} = B$
5. Take complement: $X = \bar{B}$

Key insight: The event A actually disappears from the final answer because $\bar{A} + A = S$, meaning the condition holds regardless of whether A occurs or not. The equation essentially states that X is the complement of B .